INCH POUND

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DETAIL SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED, DELAY, M10, M31, M42, M72, AND M114 REMANUFACTURED METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M10, M31, M42, M72, and M114 Delay Cartridge Actuated Initiator Remanufactured Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MILITARY

MIL-A-2550 Ammunition, General Specification for

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/

AMSC N/A FSC 1377

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DEPARTMENT OF DEFENSE STANDARDS

MILITARY

MIL-STD-413	Visual Inspection Guide for Elastomeric O-rings

(Copies of these documents are available online at http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

8594862	Initiator, Cartridge Actuated, Delay, Metal Parts Assembly
8798143	Box, Fiberboard (Inner) Packing, Ammunition for Initiator, Cartridge Actuated, Delay, M10, M30A1, M31, M32A1, M42, M43A1, M72, M92, or M114
8798142	Box, Packing, Ammunition for Initiator, Cartridge Actuated, Delay, M10, M30A1, M31, M32A1, M42, M43A1, M72, M92, or M114
9216933	Box, Fiberboard (Outer) Alternative, Packing Ammunition, for Initiator, Cartridge Actuated, Delay, M10, M30AI, M31, M32A1, M42, M43A1, M72, M92, or M114

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Technical Information Branch (Code 4230), Indian Head, MD 20640-5035.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARD INSTITUTE/AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Zl.4	Sampling Procedures and Tables for Inspection by Attributes
ANSI/ASQ A8402	Quality Management and Quality Assurance - Vocabulary

(Copies of these documents are available online at http://webstore.ansi.org/ansidocstore/default.asp, or from the American Society for Quality, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the initiators shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal. The remanufactured initiator metal parts assembly shall contain a new shear pin and O-rings.

3.2 Hydrostatic pressure. The initiator chamber and cap shall withstand a hydrostatic pressure of 5500 to 6500 pounds per square inch (psi) for 15 seconds, minimum, without deformation or mechanical failure (see 6.3.1).

3.3 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component, regardless of configuration. Remanufactured exterior components shall have the existing protective finish stripped and the specified protective finish reapplied (see 6.3.2).

3.4 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using procedures in 4.5.3.

3.5 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 41 ± 5 pounds is applied at the steady rate of 0.125 inch per minute.

3.6 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled handtight.

3.6.1 Integrated plant products. When remanufacturing and loading of assembly is performed in an integrated plant, the metal parts assembly requirement may be waived and the residual magnetism requirement performed on the loaded assembly only.

3.6.2 Threaded component parts. Class 2A or 2B thread gage shall be used on component part inspection versus class 3A or 3B required on remanufactured hardware for initiators.

3.7 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following:

3.7.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.7.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

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4. VERIFICATION

4.1 Quality assurance terms and definitions. Reference shall be made to ANSI/ASC A8402 to define the quality assurance terms used.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 68° to 77°F
b. Altitude: Sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psia

4.3 Quality conformance inspection.

4.3.1 Lot.

4.3.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with ANSI/ASQ Z1.4.

4.3.1.2 Size of lot. For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 600 assemblies, maximum, provided the homogeneity requirements of inspection lots have been retained.

4.3.2 Examination. On reprocessed parts one-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4×) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristics shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.3.2.1 Examination of reclaimed parts. Examination of reuseable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be reprocessed or rejected.

4.3.2.2 Retainer, Cartridge, Drawing 8594860, covering a detail of Drawing 8594862.

Characteristic	Defect	Method of Inspection
Critical		
1	Metering slots in base missing	Visual
Major		
101	Pitch diameter of thread, max	Gage
102	Minor diameter of thread, max	Gage
103	Elongated slots, missing	Visual

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104	Thread damaged	Visual
105	Metal defective	Visual
106	Protective finish inadequate or defective	Visual
107	Surface finish on internal radius and base, improper	Visual
108	Burr	Visual
109	Foreign matter present	Visual

Minor
None defined

4.3.2.3 Chamber, Initiator, Drawing 8594857 covering a detail of Drawing 8594862.

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Pitch diameter of large thread, min.	Gage
102	Major diameter of large thread, min.	Gage
103	Pitch diameter of small thread, min.	Gage
104	Major diameter of small thread, min.	Gage
105	Thread damaged	Visual
106	Metal defective	Visual
107	Protective finish inadequate or defective	Visual
108	Surface finish, end of gas port, improper	Visual
109	Burr	Visual
110	Foreign matter present	Visual
Minor		
None defined		

4.3.2.4 Cap, Drawing 8595254 covering a detail of Drawing 8594862.

Characteristic	Defect	Method of Inspection
Critical		
1	Metal defective in mounting flange surface	Visual
Major		
101	Pitch diameter of thread, max.	Gage
102	Minor diameter of thread, max.	Gage
103	Thread damaged	Visual
104	Metal defective	Visual
105	Protective finish inadequate or defective	Visual
106	Burr	Visual
107	Foreign matter present	Visual
Minor		
None defined		

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4.3.2.5 Pin, Shear, Drawing 8594745, covering a detail of Drawing 8594862.

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Diameter, max.	Gage
102	Total length, min.	Measure
103	Metal defective	
104	Protective finish inadequate or defective	Visual
105	Surface finish improper	Visual
106	Burr	Visual
107	Foreign matter present	Visual
Minor		
201	Total length, max.	Gage
202	Profile of end	Gage

4.3.2.6 Housing, Firing Pin, Drawing 8594858 covering a detail of Drawing 8594862.

Characteristic	Defect	Method of Inspection
Critical		
C1	Surface finish of cavity	Gage
Major		
101	Pitch diameter of internal thread, max	Gage
102	Minor diameter of internal thread, max	Gage
103	Pitch diameter of external thread, min	Gage
104	Major diameter of external thread, min	Gage
105	Thread damaged	Visual
106	Metal defective	visual
107	Protective finish inadequate or defective	visual
108	Surface finish of cavity chamfer improper	Visual
109	Burr	Visual
110	Foreign matter present	Visual
Minor		
None defined		

4.3.2.7 Pin, firing, Drawing 8594744 covering a detail of Drawing 8594862.

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Metal defective	Visual

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102	Protective finish inadequate or defective	Visual
103	Surface finish of body and striker tip, improper	Visual
104	Burr	Visual
105	Foreign matter present	Visual
Minor		
None defined		

4.3.2.8 Assembly, less Shear Pin, Cap, Cartridge Retainer and Chamber, Drawing 8594862, less Drawings 8594745, 8595254, 8594860, and 8594857.

Characteristic	Defect	Method of Inspection
Critical		
1	Protrusion of firing pin	Gage
2	O-ring missing	Visual
Major		
None defined		
Minor		
201	O-ring not lubricated	Visual

4.3.2.9 Assembly, less Cap, Chamber, Cartridge Retainer, Plug Machine Screw and Cap Machine Screw, Drawing 8594862 less Drawings 8595254, 8595857, 8594860, NAS845-3 and NAS832-4.

Characteristic	Defect	Method of Inspection
Critical		
1	Shear pin not engaged in firing pin	Visual/Manual
2	Shear pin set screw missing	Visual
Major		
101	O-ring missing or not lubricated	Visual
102	Thread damaged	Visual
103	Foreign matter present	Visual
Minor		
None defined		

4.3.2.10 Assembly, Drawing 8594862.

Characteristic	Defect	Method of Inspection
Critical		
1	Any part missing, except as otherwise classified	Visual
Major		
None defined		

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Minor		
201	Protective cap or plug missing	Visual
202	Protective finish inadequate or defective	Visual

4.3.2.11 Box, Fiberboard (inner) Packing Ammunition, Drawing 8798143.

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Box damaged	Visual
203	Marking missing, incorrect, or illegible	Visual
204	Cut, damaged or improperly sealed bag	Visual

4.3.2.12 Box, Packing, Ammunition, Drawing 8798142.

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Visual/Manual
202	Strapping missing, loose, broken, or improperly assembled	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.3.2.13 Box, Fiberboard (outer) Alternative, Packing Ammunition, Drawing 9216933.

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual/Visual

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202	Hardware missing, loose, broken, or improperly engaged	Visual
203	Strapping missing, loose, broken, or improperly engaged	Visual
204	Contents of box exposed	Visual
205	Box damaged	Visual
206	Marking missing, incorrect or illegible	Visual

4.4 Inspection testing. The following tests shall be performed in accordance with the provisions of 4.5.

4.4.1 Hydrostatic pressure. Each chamber and cap shall be subjected to the hydrostatic pressure test. Failure of any cap or chamber subassembly to comply with the requirement of 3.2 shall be cause for rejection of the subassembly.

4.4.2 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.4.3 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism specified in 4.5.3. Failure of any unit in the inspection sample to comply with the requirement of 3.4 shall be cause for rejection of the assembly.

4.4.4 Shear pin strength. Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any pin to comply with the requirement of 3.5 shall be cause for rejection of the lot.

4.5 Test methods and procedures.

4.5.1 Hydrostatic pressure. Prior to the application of protective finish, the cap and chamber shall be mounted in the device, using a fixture for properly sealing all openings and subjected to the internal hydrostatic pressure to determine compliance with the requirement of 3.2.

4.5.2 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.5.3 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.5.4 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with 3.5. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.6 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The assembly will be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems. Since the initiator was developed specifically for use in military aircraft, there is no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following;

a. Title, number, and date of this specification
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1)
c. The responsible inspection activity (see 4.1)
d. Any special inspection conditions (see 4.2)
e. Level of marking and packing (see 5.1 and 6.5)

6.3 Definitions.

6.3.1 Mechanical failure. Mechanical failure is any deformation or breakage of apart, the occurrence of which is other than a design function.

6.3.2 Exterior components. Remanufactured exterior components are: cap, firing pin housing, and chamber.

6.4 Subject term (keyword) listing

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.5 Contract packaging, packing and marking. The following requirements should be specified in the contract.

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6.5.1 Packing and marking should Level A or B, as specified by the contracting activity.

6.5.1.1 Level A. Level A should be in accordance with Drawings 8798143 and 8798142 or SPI F00-138-3770.

6.5.1.2 Level B. Level B should be in accordance with Drawings 8798143 and 9216933 or SPI F00-138-3770.

6.5.1.3 Lot numbers. Lot numbers should be in accordance with MIL-STD-1168.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue because of the extensiveness of the changes. This revision was authorized by ECP number 045220I057.

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